

Операционные системы

Командная оболочка Midnight Commander

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1. Цели и задачи работы
2. Процесс выполнения лабораторной работы
3. Выводы по проделанной работе

1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

1.2 Задачи лабораторной работы

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. 2. Процесс выполнения лабораторной работы

2.1 Работа с Midnight Commander

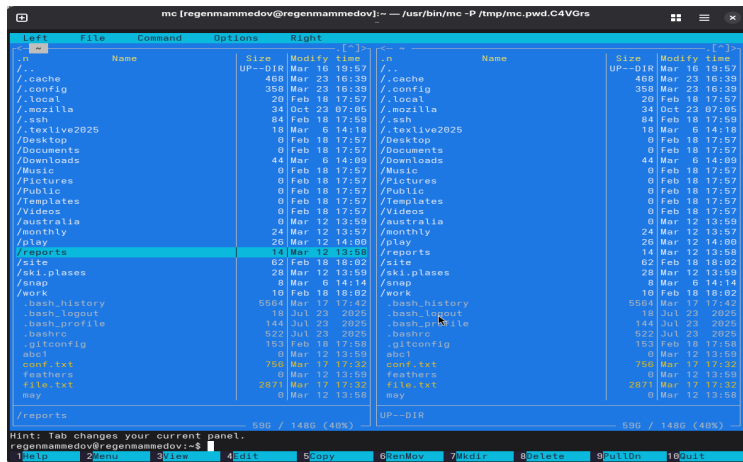


Рисунок 1: Запуск mc

2.2 Работа с Midnight Commander

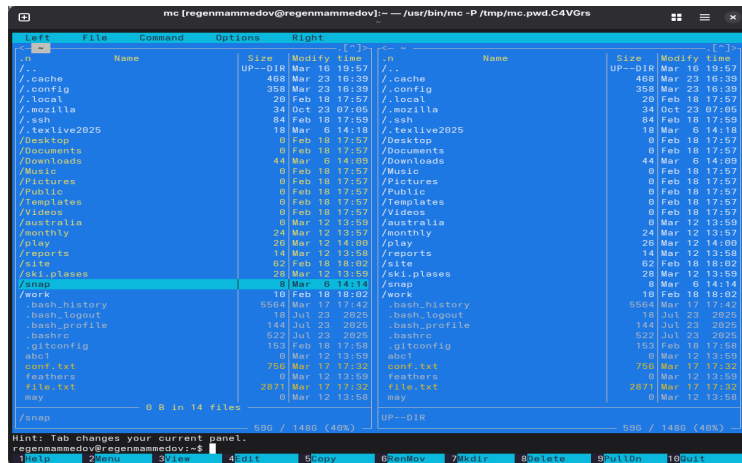


Рисунок 2: Выделение

2.3 Работа с Midnight Commander

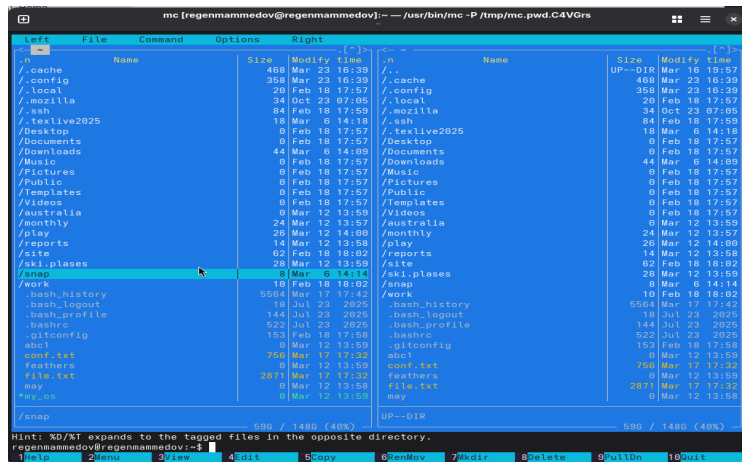


Рисунок 3: Отмена

2.4 Работа с Midnight Commander

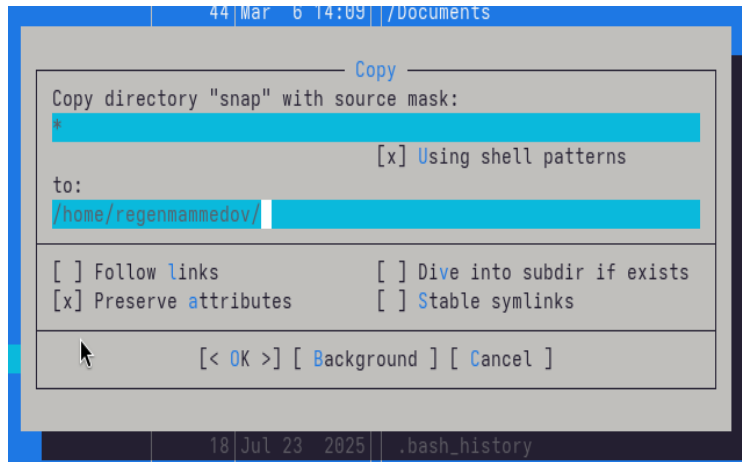


Рисунок 4: Копирование

2.5 Работа с Midnight Commander

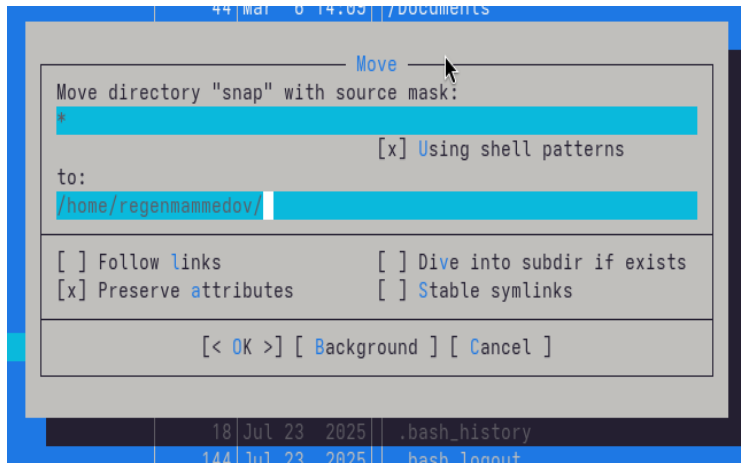


Рисунок 5: Перемещение

2.6 Работа с Midnight Commander

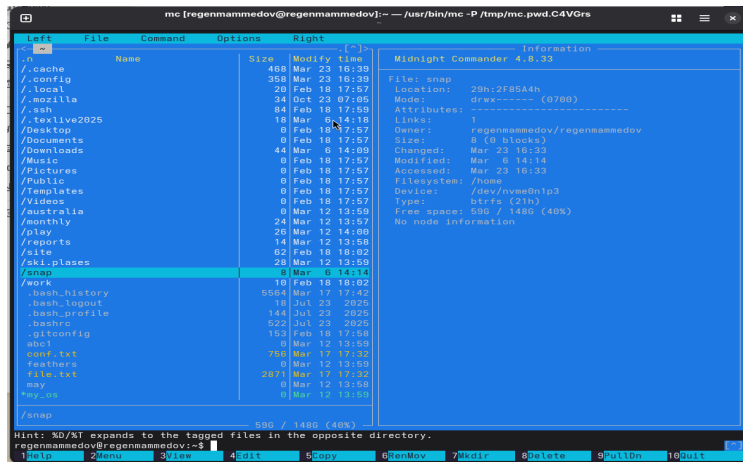


Рисунок 6: Информация

2.7 Работа с Midnight Commander

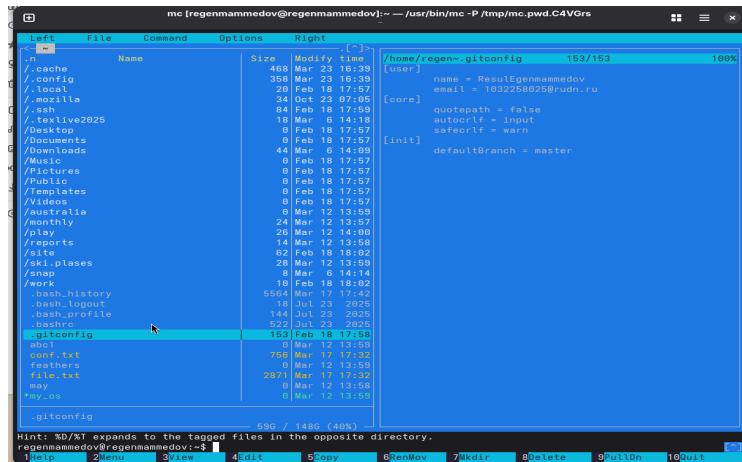


Рисунок 7: Быстрый просмотр

2.8 Работа с Midnight Commander

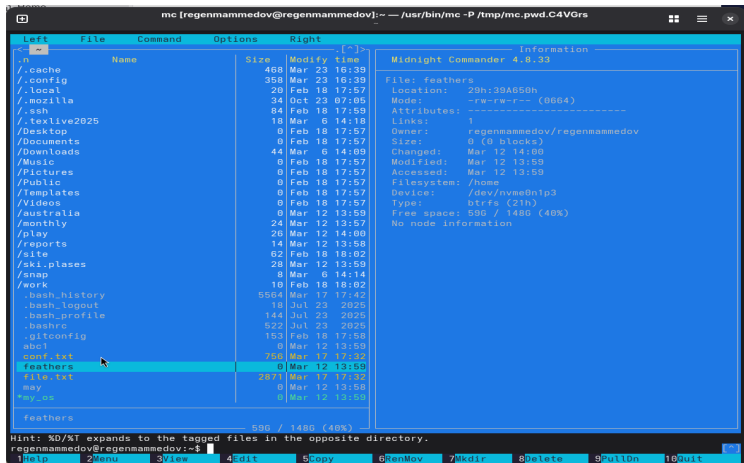


Рисунок 8: Информация

2.9 Работа с Midnight Commander

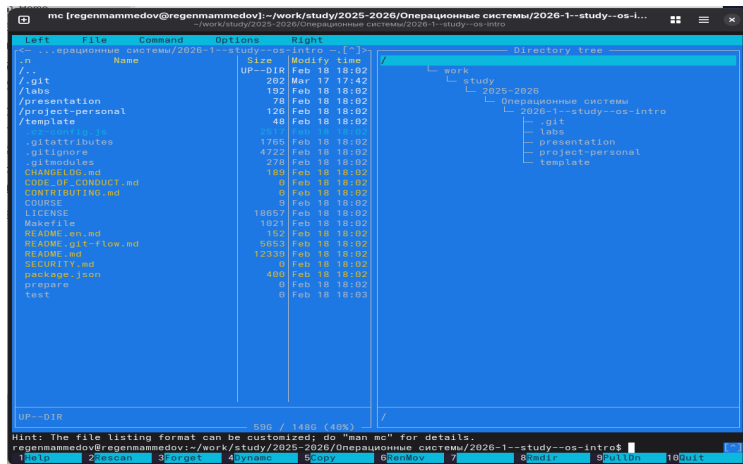
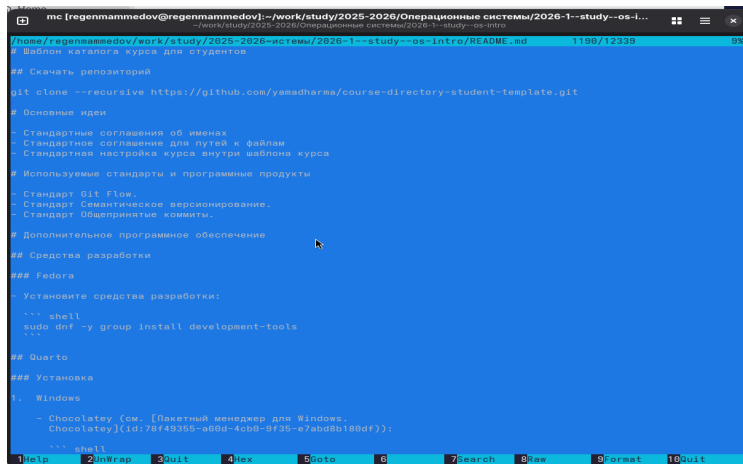


Рисунок 9: Дерево каталогов

2.10 Работа с Midnight Commander



The screenshot shows the Midnight Commander (mc) interface. The title bar indicates the current directory is `~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro`. The main window displays the contents of the file `/home/regenmammedov/work/study/2025-2026-истемы/2026-1--study--os-intro/README.md`, which is 1198/12339 bytes in size. The file content is as follows:

```
# Шаблон каталога курса для студентов

## Скачать репозиторий

git clone --recursive https://github.com/yamadharma/course-directory-student-template.git

# Основные идеи

- Стандартные соглашения об именах
- Стандартное соглашение для путей к файлам
- Стандартная настройка курса внутри шаблона курса

# Используемые стандарты и программные продукты

- Стандарт Git Flow.
- Стандарт Семантическое версионирование.
- Стандарт Общепринятые коммиты.

# Дополнительное программное обеспечение

## Средства разработки

### Fedora

- Установите средства разработки:

  ``` shell
 sudo dnf -y group install development-tools
  ```

### Quarto

### Установка

1. Windows

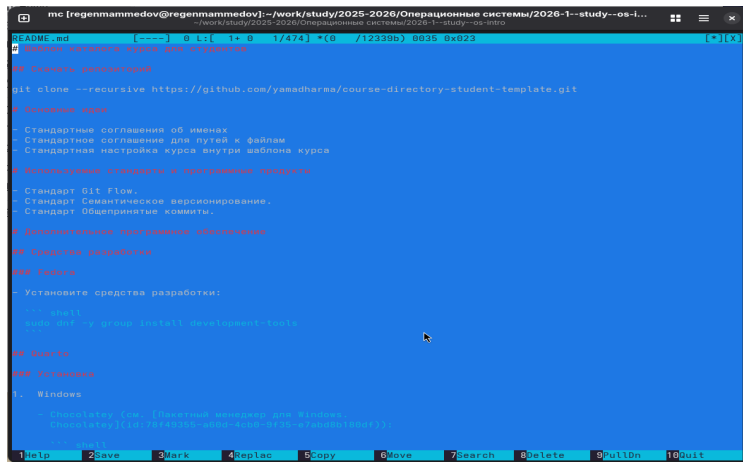
  - Chocolatey (см. [Пакетный менеджер для Windows. Chocolatey](id:78f49355-a60d-4cb0-9f35-e7abd8b180df)):

    ``` shell
```

At the bottom of the window, there is a toolbar with buttons numbered 1 to 10: 1Help, 2UnWrap, 3Quit, 4lex, 5Coto, 6, 7Search, 8Rew, 9Format, 10Quit.

Рисунок 10: Просмотр содержимого текстового файла

## 2.11 Работа с Midnight Commander



The screenshot shows the Midnight Commander (mc) interface. The title bar indicates the current directory is `~/work/study/2025-2026/Операционные системы/2026-1--study--os-i...`. The main window displays the content of the `README.md` file. The text is as follows:

```
README.md [----] 0 L: [1+ 0 1/474] *(0 /12339b) 0035 0x023 [*][x]
Базовый каталог курса для студентов

Создать репозиторий

git clone --recursive https://github.com/yamadharma/course-directory-student-template.git

Основные идеи

- Стандартные соглашения об именах
- Стандартное соглашение для путей к файлам
- Стандартная настройка курса внутри шаблона курса

Используемые стандарты и программные продукты

- Стандарт Git Flow.
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- Стандарт Общепринятые коммиты.

Дополнительное программное обеспечение

Средства разработки

Fedora

- Установите средства разработки:

 ... shell
 sudo dnf -y group install development-tools

Ubuntu

Windows

- Chocolatey (см. [Пакетный менеджер для Windows.
 Chocolatey](https://chocolatey.org/packages/chocolatey)).

 ... shell
```

At the bottom of the window, there is a toolbar with numbered buttons: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

## 2.12 Работа с Midnight Commander

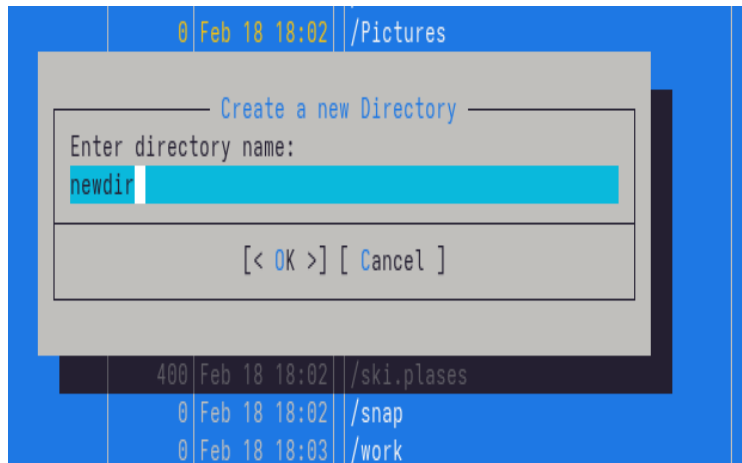


Рисунок 12: Создание каталога

## 2.13 Работа с Midnight Commander

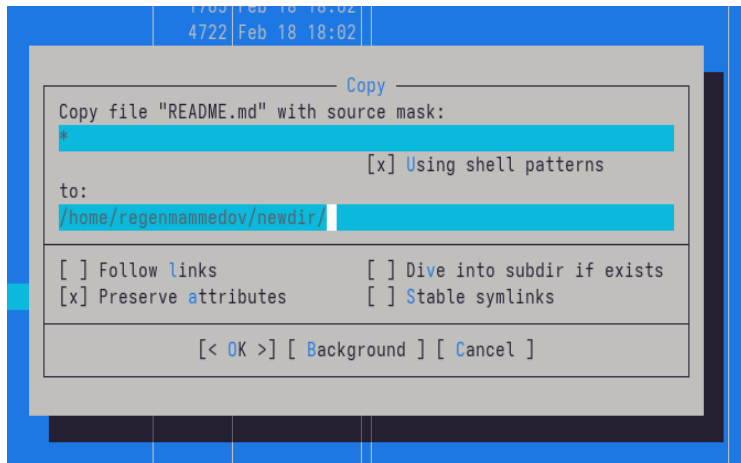


Рисунок 13: Копирование в файлов в созданный каталог

## 2.14 Работа с Midnight Commander

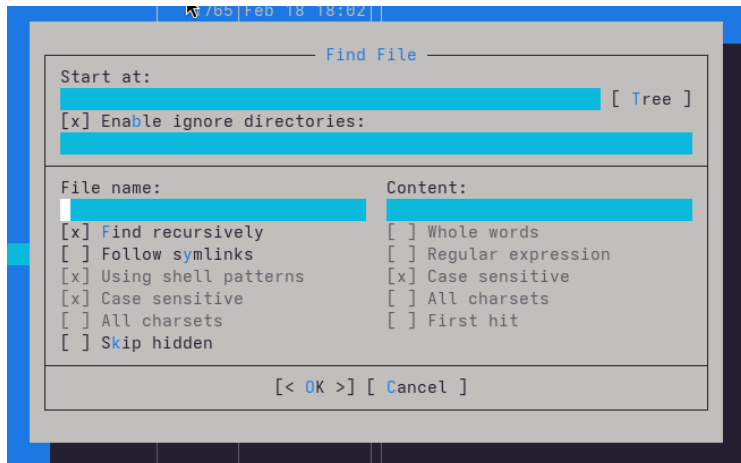


Рисунок 14: Поиск файлов

## 2.15 Работа с Midnight Commander

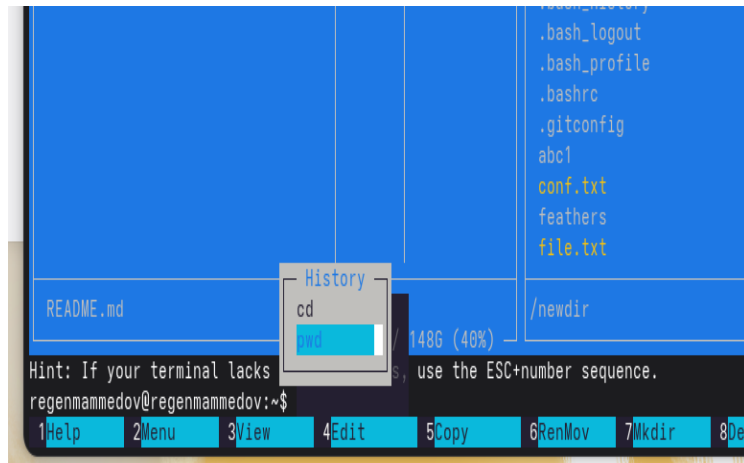


Рисунок 15: История команд

## 2.16 Работа с Midnight Commander

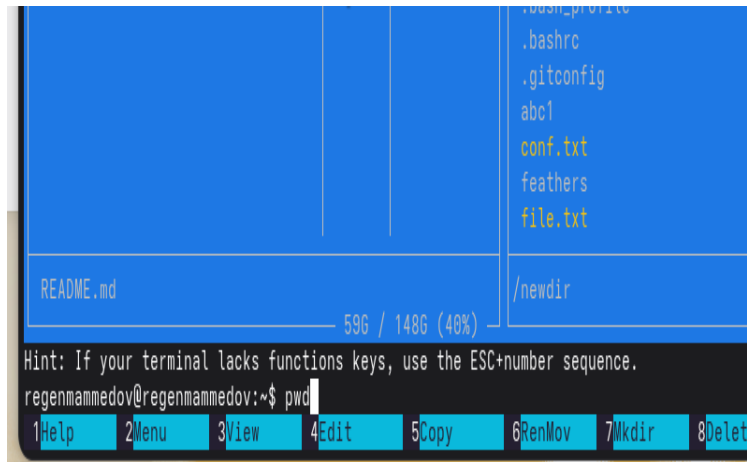
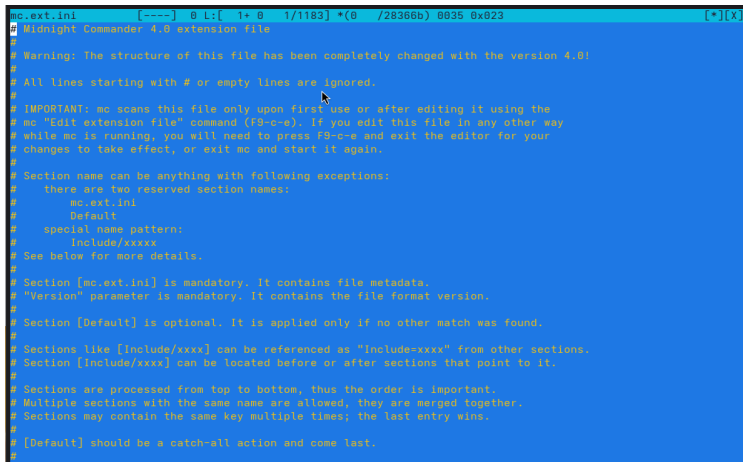


Рисунок 16: Переход в домашний каталог

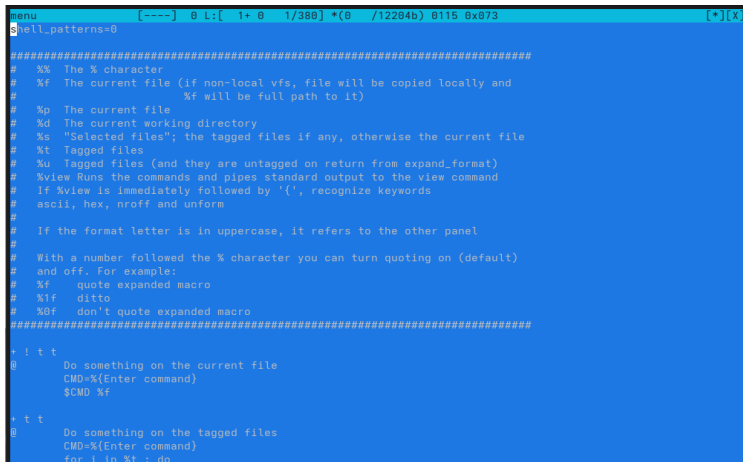
## 2.17 Работа с Midnight Commander



```
mc.ext.ini [---] 0 L: 1+ 0 1/1183] *(0 /28366b) 0035 0x023 [*][X]
Midnight Commander 4.0 extension file
#
Warning: The structure of this file has been completely changed with the version 4.0!
#
All lines starting with # or empty lines are ignored.
#
IMPORTANT: mc scans this file only upon first use or after editing it using the
mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
while mc is running, you will need to press F9-c-e and exit the editor for your
changes to take effect, or exit mc and start it again.
#
Section name can be anything with following exceptions:
there are two reserved section names:
mc.ext.ini
Default
special name pattern:
Include/xxxx
See below for more details.
#
Section [mc.ext.ini] is mandatory. It contains file metadata.
"Version" parameter is mandatory. It contains the file format version.
#
Section [Default] is optional. It is applied only if no other match was found.
#
Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
Section [include/xxxx] can be located before or after sections that point to it.
#
Sections are processed from top to bottom, thus the order is important.
Multiple sections with the same name are allowed, they are merged together.
Sections may contain the same key multiple times; the last entry wins.
#
[Default] should be a catch-all action and come last.
```

Рисунок 17: Просмотр файла расширений

## 2.18 Работа с Midnight Commander

A screenshot of the Midnight Commander file menu. The window title bar shows 'menu [----] 0 L:[ 1+ 0 1/380] \*(0 /12204b) 0115 0x073 [+][X]'. The menu content is as follows:

```
shell_patterns=0

#####
%% The % character
%f The current file (if non-local vfs, file will be copied locally and
%f will be full path to it)
%p The current file
%d The current working directory
%s "Selected files"; the tagged files if any, otherwise the current file
%t Tagged files
%u Tagged files (and they are untagged on return from expand_format)
%view Runs the commands and pipes standard output to the view command
If %view is immediately followed by '{', recognize keywords
ascii, hex, proff and unform
#
If the format letter is in uppercase, it refers to the other panel
#
With a number followed the % character you can turn quoting on (default)
and off. For example:
%f quote expanded macro
%if ditto
%0f don't quote expanded macro
#####

+ ! t t
0 Do something on the current file
 CMD=%(Enter command)
 $CMD %f

+ t t
0 Do something on the tagged files
 CMD=%(Enter command)
 for i in %t ; do
```

Рисунок 18: Просмотр файла меню

## 2.19 Работа с Midnight Commander

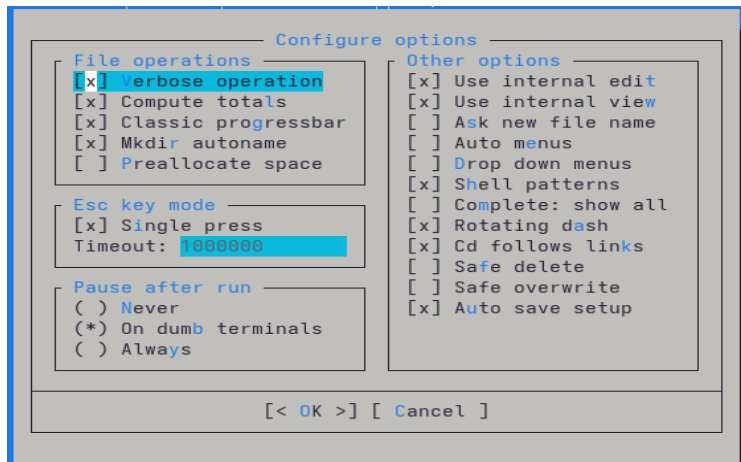


Рисунок 19: Конфигурация

## 2.20 Работа с Midnight Commander

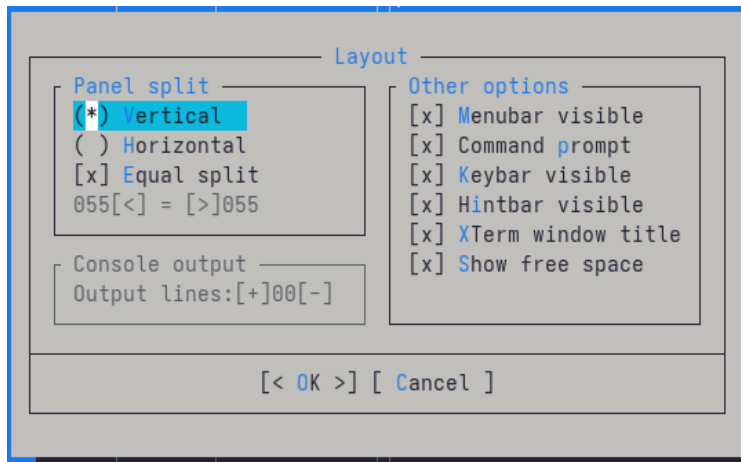


Рисунок 20: Внешний вид

## 2.21 Работа с Midnight Commander

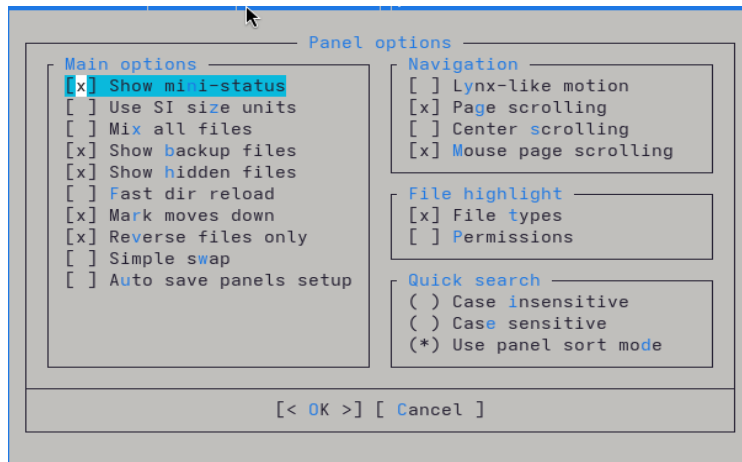


Рисунок 21: Настройки панелей

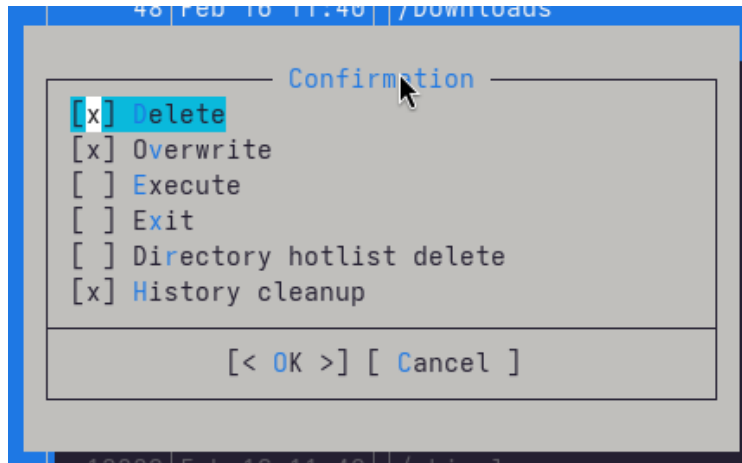


Рисунок 22: Подтверждение

## 2.23 Работа с Midnight Commander

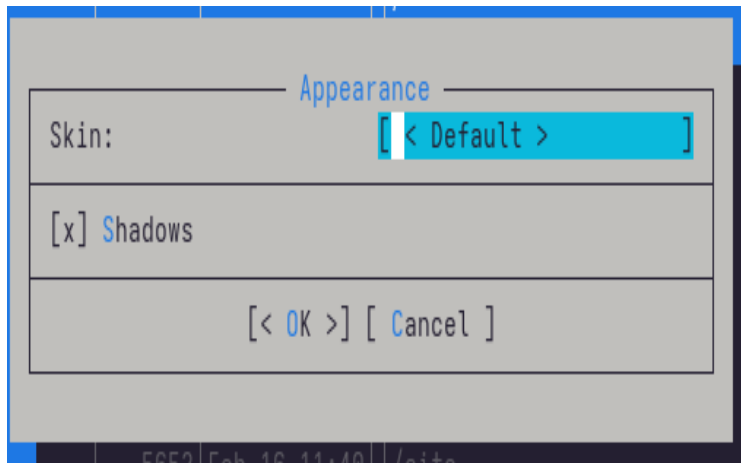


Рисунок 23: Оформление

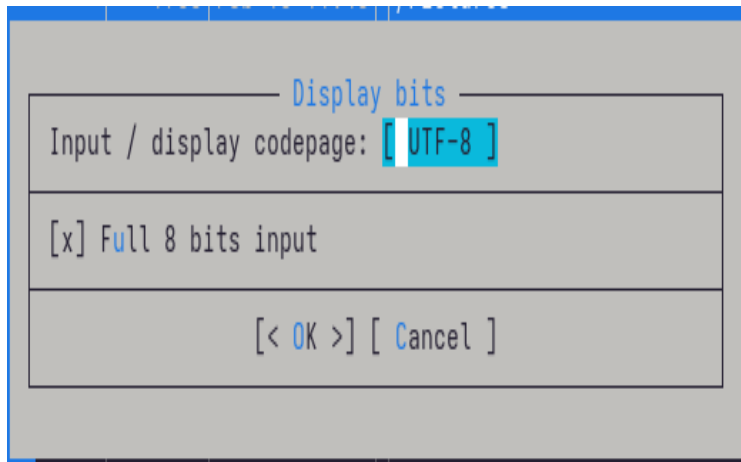


Рисунок 24: Кодировка символов

## 2.25 Работа с Midnight Commander

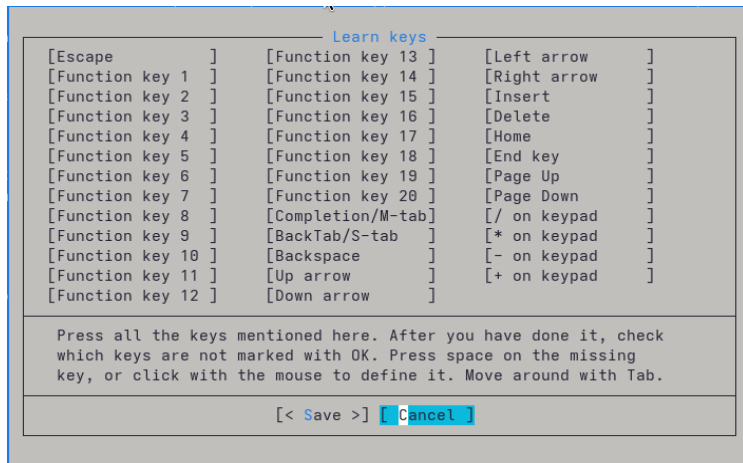


Рисунок 25: Распознавание клавиш

## 2.26 Работа с редактором Midnight Commander

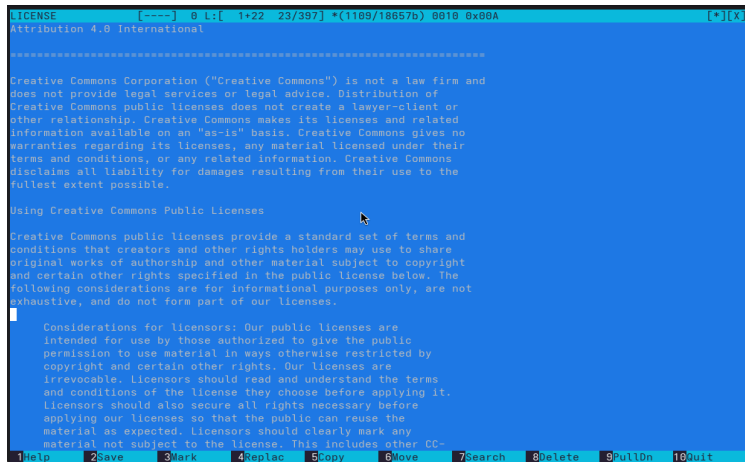
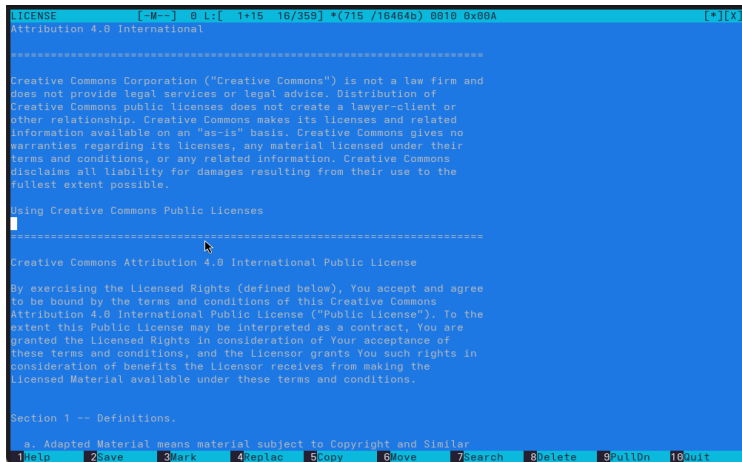


Рисунок 26: Файл с текстом

## 2.27 Работа с редактором Midnight Commander



```
LICENSE [-M--] 8 L:[1+15 16/359] *(715 /16464b) 0010 0x00A [*][x]
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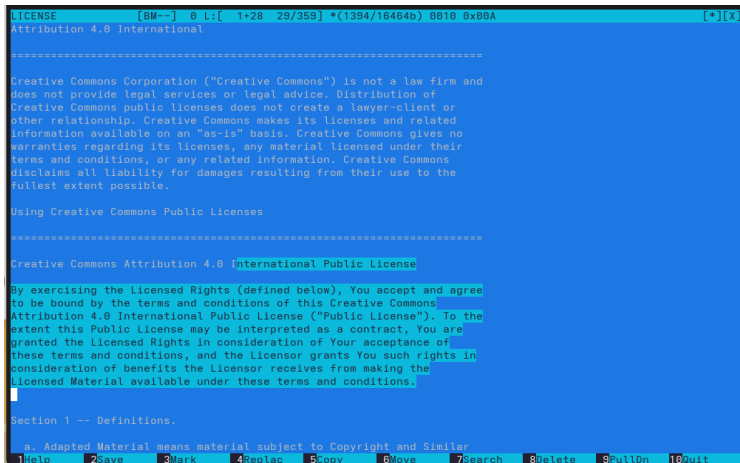
Section 1 -- Definitions.

a. Adapted Material means material subject to Copyright and Similar

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 27: Файл с текстом

## 2.28 Работа с редактором Midnight Commander



```
LICENSE [BM--] 0 L:[1+28 29/359] *(1394/16464b) 0010 0x00A [*][X]
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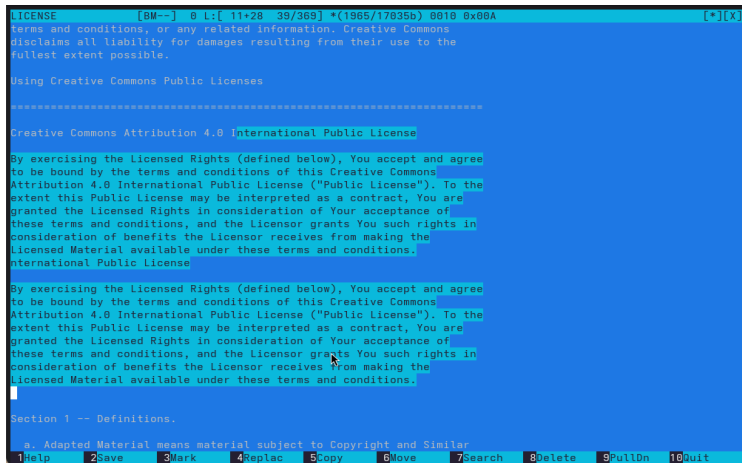
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1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullOn 10Quit
```

Рисунок 28: Копирование фрагмента

## 2.29 Работа с редактором Midnight Commander



```
LICENSE [BM--] 0 L:[11+28 39/369] *(1965/17035b) 0010 0x00A [*][X]
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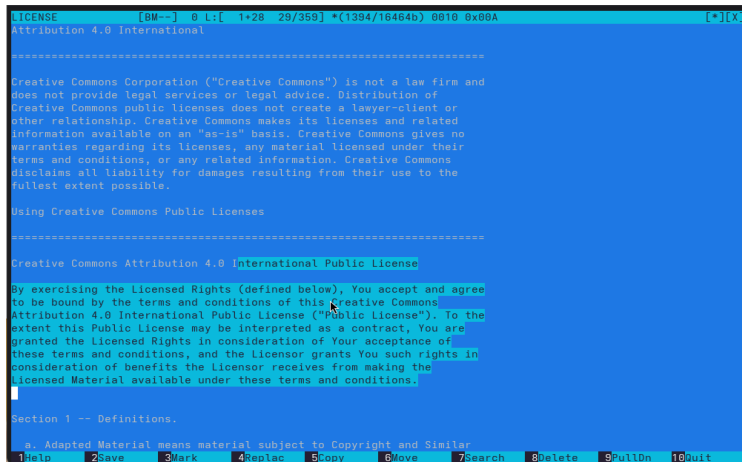
Section 1 -- Definitions.

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1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 29: Сохранение

## 2.30 Работа с редактором Midnight Commander



```
LICENSE [BM--] 8 L:[1+28 29/359] *(1394/16464b) 0010 0x00A [*][X]
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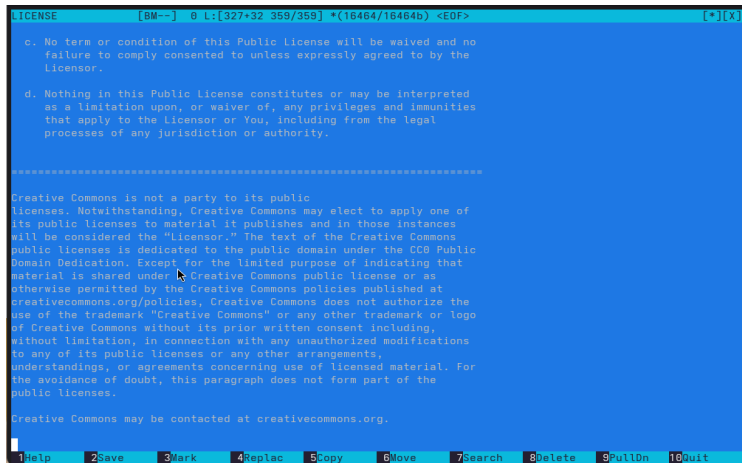
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1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 30: Отмена

## 2.31 Работа с редактором Midnight Commander



```
LICENSE [BM--] 0 L:[327+32 359/359] *(16464/16464b) <EOF> [*][X]

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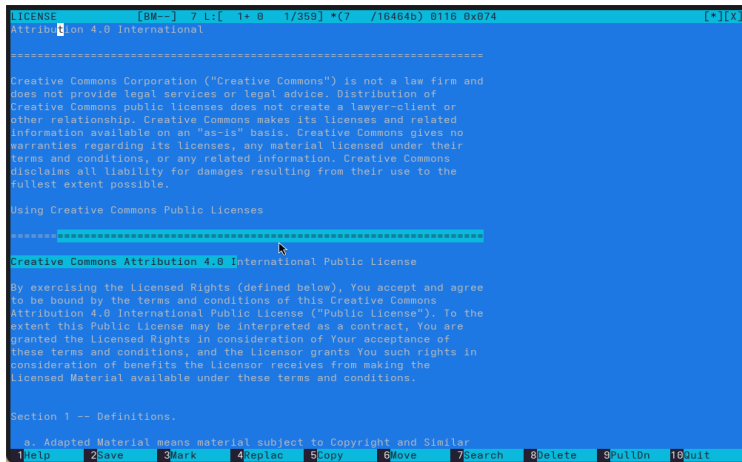
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1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 31: Переход в конец файла

## 2.32 Работа с редактором Midnight Commander



```
LICENSE [BM--] 7 L:[1+ 0 1/359] *(7 /16464b) 0116 0x074 [*][X]
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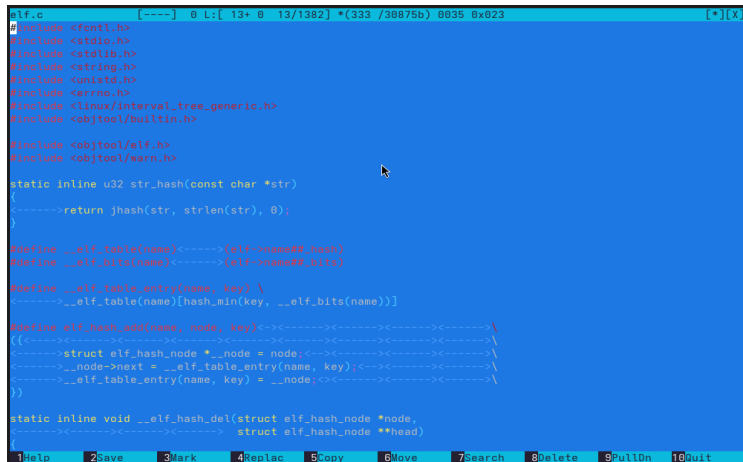
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1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 32: Переход в начало файла

## 2.33 Работа с редактором Midnight Commander



```
elf.c [----] 0 L: [13+ 0 13/1382] *(333 /30875b) 0035 0x023 [*][X]
#include <fcntl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <errno.h>
#include <linux/interval_tree_generic.h>
#include <objtool/builtin.h>

#include <objtool/elf.h>
#include <objtool/warn.h>

static inline u32 str_hash(const char *str)
{
 ----->return jhash(str, strlen(str), 0);
}

#define __elf_table(name)----->{elf->name##_hash}
#define __elf_bits(name)----->{elf->name##_bits}

#define __elf_table_entry(name, key) \
 ----->__elf_table(name)[hash_min(key, __elf_bits(name))]

#define elf_hash_add(name, node, key)----->----->----->----->\
 ({----->----->----->----->----->\
 ----->struct elf_hash_node *__node = node;----->----->----->----->\
 ----->__node->next = __elf_table_entry(name, key);----->----->----->----->\
 ----->__elf_table_entry(name, key) = __node;----->----->----->----->\
 })

static inline void __elf_hash_del(struct elf_hash_node *node,
 ----->----->----->-----> struct elf_hash_node **head)
```

Рисунок 33: Файл с программой

## 2.34 Работа с редактором Midnight Commander

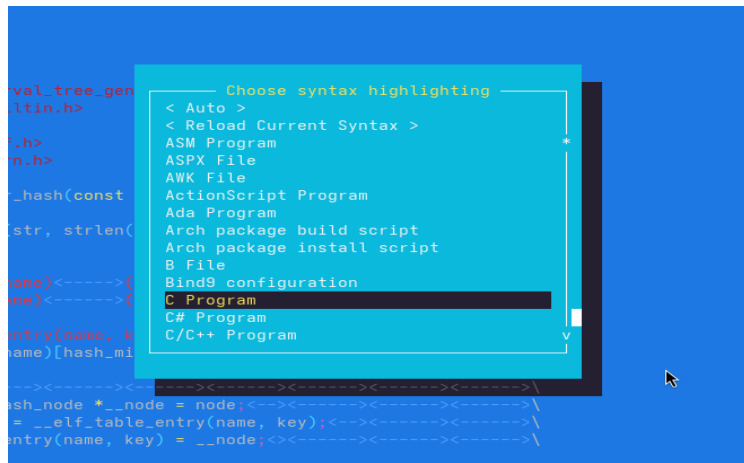


Рисунок 34: Цветовыделение синтаксиса

### 3. 3. Выводы по проделанной работе

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В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.